

2)

$$\left(\frac{n^2-3}{n^2+2}\right)^{n^2} = (n^2-3)^{n^2} \cdot [(n^2+2)^{n^2}]^{-1} = e^{-3} \cdot e^{-2} = e^{-5}$$

5)

$$a_n = \frac{(2n)!}{n^{n/2}} = \frac{(2n)!}{\sqrt{n^n}} = +\infty \text{ perche infiniti}$$

6)

$$a_n = \frac{n^8 + n! + n(-10)^n}{n^2 2^n + n^{10} - 3^n} \sim \frac{n!}{n^2 2^n} = +\infty \text{ perche infiniti}$$

8)

$$a_n = \frac{3^n - 7^n}{4^n - n 2^n} \sim \frac{7^n}{4^n} = \left(\frac{-7}{4}\right)^n = -\infty$$

1)

$$\frac{n^{-2} 10^n + (-n)^n + \cos(3^n)}{n^{1/n} + n^7 - 7^n} \sim \frac{(-n)^n}{-7^n}$$

non ha limite
perche $(-n)^n$ cambia
continuamente segno

3)

$$a_n = \frac{n^2 + 3n^4 - 7}{(2n^2+1)(n^2-1)} = \frac{n^2 + 3n^4 - 7}{2n^4 - 2n^2 + n^2 - 1} \sim \frac{3n^4}{-2n^4} = -\frac{3}{2}$$

7)